JIANT FM · SOUND · FX · MOD

engines, effects, modulation, sequencer and songs

firmware V10.6z9 · M-VAVE FM-1 · juanjiant-bit.github.io/JIANT

ENGINES · EDIT 1 / EDIT 2

TWELVE ENGINES: ELEVEN SYNTHS AND EXT

ENGINE	EDIT 1				EDIT 2				WHAT TO KNOW
DRUM-X	MORPH	TUNE	TONE	DECAY	NOISE DRIVE	FM	ENV	FOLD	8 sounds BD SD CP CH OH TM RS CB, sides A / B: **MORPH** morphs the kit. The kit first: **KIT 3** BP FEEDBK, **X-MOD** RATE DEPTH SHAPE RAND. Then a sound (KNOB 1 the sound): SIDE PITCH PMOD · DECAY NOISE COLOR · MODE PM.TYPE DRIVE · **LAYER** (no note) FEEDBK **SEND** (CHO / DLY / REV). Last LEVELS and **STEREO** IMAGE (a knob through the kit's stereo placings). **KIT 4 KEY MAP** (V10.6r): EXTENDED (default) +1 FILLS, –1 KICK / TOM or one sound chromatic, –2 FX KIT, –3 FLAM; or CLASSIC. Cowbell: classic CMOS (V10.6q). KIT, SOUND 1–4 and HOME › BEING draw the kit as little beings that squash and shout on each hit (0.9.4).
ANALOG	WAVE	DETUNE	MIX	NOISE	CUT	RES	DRIVE	INT	SAW SQR TRI SIN PWM **SYNC RING SAW3 PLS3**; INT: osc 2 ±24 st; on SAW3 / PLS3 the NOISE knob is **INT 3**, osc 3's interval (V10.39). Its filter is per voice. **Its picture** (0.9.6): a cell whose membrane is the wave, spikes where DRIVE clips, a ring at CUT.
PHASE	WAVE	WAVE2	DCW	ENV	DETUNE	LINE	SUB	FB	Phase distortion (CZ); **FB** feeds the output back into the phase: hard edge to growl. **Its picture** (0.9.6): a snail, its shell's chambers at the bent phase, its sole the wave.
FM6	ALG	FB	MOD	RATIO	MEG	VMOD	DETUNE	SLOT	Dexed's 6 operators, 32 algorithms drawn, .syx patches (SLOT), MOD / RATIO move the modulators. **OPERATOR · OP ENV · OPERATOR 2**: each operator on the device (ratio / fixed, fine, level, its envelope, mode, detune, velocity). **SLOT B1–B32**: a 32-voice .syx sent from the web editor. **Its picture** (0.9.6): the operators as bichos wired as the algorithm, OP ENV's riding its envelope.
VOICE	VOWEL	VOWEL2	TALK	SHIFT	BUZZ	BREATH	Q	RAND	Formant voice (after klattsch): vowels, talk, breath. **WORDS** (0.8.5): LANG ES / EN / JA sings syllables, a note each (POS THEME ARTIC SHIFT · BUZZ BREATH Q LOOP); LYRIC TEXT (OCT+ on WORDS types it, KNOB 3 dice; the web editor sends it) or W1–16 words, S1–8 scat; "-" holds the vowel. **Its picture** (0.9.4): a CT head whose tongue, jaw, lips and velum sing the phoneme.
ANIMAL	MODE	PRESS	CHAOS	SIZE	SWEEP	RATE	GROUP	TEMP	Animal calls (0.8.3), MODE the source: **CMOS** the 40106 / 4093 bird circuit (TOUCH COUPLE CAP · HYST RATE GROUP FM) · **PULSE** crickets, cicadas, frogs (CLICK Q · DETUNE) · **WHSTL** birds, a Hopf whistle · **WING** mosquito, bee, fly (BUZZ DRIFT · DOPPLR) · **GLOTT** wolf, bark, roar, purr (MOUTH ROUGH · AIR). GROUP: syllables + gaps; TEMP °C speeds a cricket up (Dolbear). A note is a call. **Its picture** (0.9.4): bird ↔ insect ↔ mammal by MODE, leaned by SIZE.
MATH	CHIP	WAVE	ALGO	VAR	MOTN	GRIT	BEND	LOOP	(V10.6e: LOFI's formulas alone; its chip waves load as ANALOG) **BYTE**: ALGO VAR · MOTN GRIT BEND LOOP. **FLOAT** (floatbeat): ALGO VAR · MOTN FBK BEND LOOP; CHIP sets FULL / 4BIT / 1BIT. Every sound starts still (MOTION 0), RTRG on: in tune. 48 formulas each in BYTE and FLOAT (ALGO 0–191); MOTN past half is **SYNC**: the formula's slow bits follow the bar.
WHEEL	REG	SUB	BODY	TOP	PERC	CLICK	DRIVE	ROTOR	Drawbar organ: registrations, percussion, rotary. **Its picture** (0.9.6): nine tube worms, one a drawbar, bent by the rotor.
NOISE	MODE	COLOR	FREQ	PEAK	TRK	DENSE	DRIFT	CRUSH	Coloured and metallic noise, tracked or free. **Its picture** (0.9.6): a swarm walking by COLOR, swirling by FREQ.
STRING	EXCITE	DECAY	BRIGHT	POS	STRETCH	BODY	SIZE	BOW	Karplus-Strong string (snd5): EXCITE NOISE PLUCK PICK HAMMR STICK; POS where it is struck, STRETCH a stiff, metallic string, BODY / SIZE its box, **BOW** a bow's noise into the loop. RELEASE is its damper: the string rings after the step and dies out when let go. **FDBK**: guitar feedback (MOVE where POS was; EDIT 2 GAIN DIST SPEAKR BLOOM). NYLON, STEEL GTR, HARP, KOTO, PIANO, MARIMBA, BOWED…
ADD	MODE	MORPH	BRIGHT	PARTIAL	SHIFT	COMB	DECAY	MOTION	Additive (snd5): partials by MODE HARM ODD WIDE BELL BAR DRUM **PRIME FIFTH SUB PLATE** (par6); MORPH between spectra, PARTIAL how many, SHIFT stretches them, COMB carves notches, DECAY the highs die first, MOTION sweeps the spectrum. Presets PRIME ORG, QUINT, DRONE, GONG, TOTEM…
EXT	SOURCE	GAIN	GATE	PLAY	WIDTH	RING	BITS	RATE	(V10.54; EXT since V10.6d) SOURCE **SWARM**: a melody from the Bluetooth around (PROX first, then AIR; FAR / NEAR morph, CHAOS). Else not a synth: the USB audio input on a track of its own (an interface on USB HOST behind a powered hub, or the computer with USB IN+OUT). SOURCE IN 1 · IN 2 · 1+2 · STEREO (two tracks can take IN 1 and IN 2); PLAY **THRU** (always) or **KEYS** (only while a note plays, at its envelope: the steps gate it); RING a sine at the note (KEYS); BITS, RATE crush it. FILTER, DIST, LEVEL, PAN, sends as any track; the page shows both inputs' meters. **MOD SRC FOLLOW**: its envelope. Presets IN 1, MIC GATE, SPACE IN, TRANCE GATE, RING IN, CRUSH IN…

FILTER · EVERY ENGINE

TYPE	CUT	RES	DRV

LP · HP · BP · COMB · COMB- · MS20 · LP24 (LP24: four poles, V10.22). COMB rings tuned to the note (COMB-: hollow, odd harmonics): CUT moves it ±32 st, RES how much it rings. ANALOG and MATH: per voice; the others: on the track, following LFO / ENV DEST FLT (DRV = the track's DIST).

VOICE · PLAYING

VCE	GLD	GLMOD	PRIO

POLY · MONO · LEG · UNI; glide and its mode; LAST / LOW / HIGH priority. **8 voices** shared by the six tracks (MONO and LEGATO keep their note). **EDIT › STEREO IMAGE** (V10.49): in POLY each voice its own place across the stereo field, drawn on the page (V10.52).

DICE

EDIT + SELECT: a random factory preset of the engine, every parameter moved up to a quarter of its range. DRUM-X: both sides of every sound around the factory kit. Heard at once: OCT+ OK keeps it, OCT- CANCEL brings the sound back. SEQ / LFO / FX + SELECT: the sequence, the LFO, the effects, the same way. HOME + SELECT (or HOME twice on HOME): **GEN**, the song generator, HOME's last three pages: GENRE (**EXPLORE**, where it opens: no genre, every rule broken; RANDOM; 43 genres; PRESETS too) · SEED · ENERGY · FORM; DRUMS · BASS · CHORDS · ARP · LEAD amounts; COMPLEX · SWING · HUMAN (SELECT: the next page). OCT+ makes the piece (again: a new seed), OCT- back to ECO. GLO + G4 undoes.

MODULATION

ENV · LFO

ENV

ATK	DEC	SUS	REL

ENV DEST

FLT	PIT	SHP	LOOP

LOOP: held, the envelope cycles from the sustain: an ADSR-shaped LFO. **ENV 2** (V10.22): a second ADSR, MOD SRC ENV2 on any destination.

LFO

RATE	WAVE	MODE	FADE

SIN TRI SAW SQR S&H **DRIFT** (V10.36: a smooth random wander; at 4BAR–16BAR pads breathe) · RATE up to 16BAR · MODE TRIG / FREE · LFO 2: PHASE

LFO DEST

PIT	FLT	SHP	AMP

MOD MATRIX · 4 SLOTS A TRACK

SLOT	SRC	DST	AMT

SOURCES

LFO · ENV · VEL · KEY · RAND · MODW · AT · EXPR · M1 · M2 · M3 · M4 · STEP · NOISE · PROX · ENV2 · FOLLOW

DESTINATIONS

PITCH · CUT · SHP · AMP · PAN · DIST · CHO · DLY · REV · RATE · VIB · E1…E8 · CLIP · PNCH · D.TIM · D.FB · D.TON · D.MIX · R.SIZ · R.DMP · C.RAT · C.DEP

E1…E8: the engine's own EDIT knobs. CLIP / PNCH: the master. D. R. C.: the DELAY, REVERB and CHORUS buses (a STEP source on D.TIM sequences the delay time). **PROX**: how close the Bluetooth phone is (or AIR: a hand); MENU › SYSTEM › BLUETOOTH, OCT+: FAR / NEAR, MIN MAX SPD. **FOLLOW**: the audio input's envelope (an input opens a filter, ducks a pad).

MSEQ · MACROS

MSEQ · A CV SEQUENCER PER TRACK

STEP	LVL	LEN	SLEW

16 levels, DIV on MSEQ 2, in time with the transport. Use it from MOD, SRC STEP.

MACROS M1–M4

Hold LFO: the knobs are M1–M4, latched until LFO again. Each session starts with 2 random routes a track; F3 rolls new ones, G3 clears. By hand: MOD, SRC M1…M4. Saved with the project.

SEQUENCER

64 steps a track, up to 4 notes a step, piano roll and drum grid. One chain (V10.6): STEP (the roll / grid, K1 all 64 steps) · LENGTH · TRACK · EUCLID · MODES · MUTATE · PLAYHEADS · STRETCH · COMP · COMP 2 · AUTOMATION · AUTO LIST; SELECT walks, SEQ jumps to the next group, PRESETS loads a pattern. **Hold a step + turn a knob**: a parameter lock. STEP K4 **VEL** (V10.8): 1–127, then ACC, SLD, A+S. Step recording: hold a chord + KNOB 1 its length; + ARP: arpeggiated (K1 rate, K2 order, K3 oct). COMP / COMP 2: two components a step, each ACTION · LENGTH · CONDITION · VALUE: ARP, SEQ, OCT+, OCT–, RATCH, STUT, MUTE, TRANS, RAND, LOCK, **RAMP** (V10.6v, JUMP's place: the step's lock glides there over 1–32 steps or 1–16 bars, LIN EXP LOG S STEPS ZIGZAG), SOUND (a kit's SHORT / GHOST / FLAM), BEND, SHIFT, CHORD; CONDITION ALWAYS, 25 / 50 / 75 %, 1ST, FILL, PRE, NEI (and their !), N:M, ELSE (C2: MONO). REC keeps lengths, legato (SLIDE) and velocity; GATE 100 % for new tracks (V10.6z4). TRACK: PITCH, CHANCE thins the whole track, HUMAN one amount (velocity, timing, gate, decay / release), REC Q. STRETCH / REPEAT act as the knob turns; ROTATE reads the sequence from another step; NUDGE in red (OCT+) moves the cursor step off the grid in 64ths (REC Q 0 % records where you played, to the 64th). **EUCLID** (V10.6; V10.16 VARY in four turns A A' A'' B, and **MUTATE**: EVERY 1–16 bars, AMOUNT, ORBIT / DRIFT / PULSE, GROOVE / RHYTHM / FEEL / ALL, OCT+ now, OCT– the anchor): HITS, STEPS 2–64, ROTATE write the track's pattern live; K4 LANE (DRUM-X: each sound its own; V10.7 **ALL**, the default: K1 DENSITY 1–8 writes a whole kit, a lane picked writes only that lane) or VARY; MODES adds ACCENT GHOST RATCHET VARY (DRUM-X) or MOTION HARMONY OCTAVE SLIDE; notes in the track's SCL, else the key the other tracks fit (FROM Tn), else NEW KEY; OCT+ a new variation, OCT– the one before. REC + knobs while playing: automation (64 records a section, MOTION page plays / clears them, AUTO LIST edits them). EUCLID starts at 0 and transforms the track's own sequence, giving it back as it was at rest (V10.28–V10.31). STRETCH › NUDGE: OCT– the whole sequence by steps, OCT+ the cursor step in 64ths (V10.25). Longer than 16 steps, the roll follows the playhead (V10.47).

LENGTH · TRACK

LEN	DIV	SWG	GATE
PIT	CHN	HUM	RECQ

SCALE · ARP

ROOT and SCALE are every melodic track's: CHR MAJ MIN DOR MIX PEN MPEN HARM PHRY LYD LOC MEL BLUES WHOLE DIMHW DIMWH. QNT snaps keys or the sequence, **ALL TR** (SCL KNOB 4) moves every sequence along the scale. **CHORD**: a key plays a chord (TRIAD 7TH ADD9 SUS2 SUS4 in key, MAJ MIN 7 MAJ7 MIN7 POWER), VOICING LEAD leads each to the next, STRUM as a harp. **SCENE** (V10.24): each track a role (KEEP CHORD ARP BASS LEAD DRUM; BASS / LEAD a PATTERN, DRUM a STYLE or GEN) and one key plays them all; OCT+ dices the roles, REC records each.

ARP MODES

OFF UP DN UPDN RND ORD REPEAT DNUP UP+8 CONV DIVG PINKY THUMB WALK CHORD **TRNS** (the keys transpose the sequence)

SONGS

6 projects × 4 sections (A–D) (V10.6p; 8 before), a song of up to 16 rows by bars (SAVE › PROJECT › ARRANGE, OCT+ adds a row, OCT– deletes one, PLAY plays; END: STOP, x2–x8, LOOP), SLOOP style. **TIMELINE** (V10.6o, SAVE's first): the song as blocks, the row picked opened by track: K1 row K2 part K3 bars K4 the track's **VAR** (MUTE THIN FILL HALF REV SHIFT OCT± VAR 1–4), F3–D4 mutes, F4–B4 **FROM** A–D, PLAY loops the row. **PARTS**: the four parts, OCT+ loops one. **CLIPS** (V10.6s): 6 tracks × 4 parts, OCT+ on the next bar, REC writes the rows. A sound changed while the song plays stays (V10.6z8; EDIT › RECALL). Each row can hold a **scene**: mutes, DRUM-X group mutes, macros and MIDI punch-ins. SAVE held plays sections live on the next bar, stores them, records the song (SONG REC). Each section keeps its sounds, steps, motion, kit and its SLICER.

EFFECTS

PER TRACK

FX · SENDS

DST	CHO	DLY	REV

DIST

DST	TYPE	TONE	-

SOFT · HARD · FOLD · CRUSH · RECT.

AUTO PAN

DEPTH	RATE	SYNC	WAVE

The track's pan swung L–R around its PAN. SYNC **BEAT**: RATE 4 BARS … 1/32, in time with the bar; **FREE**: in Hz. SIN · TRI · SAW · SQR · S&H, never a click.

SLICER · ONE ON A BUS

SLCR	PAT	RATE	DEPTH

OFF · GATE · STUT, 16 patterns, 1/8 … 32T. **SL BUS**: T1–T6 and **FX** (the delay / reverb / chorus returns) ON / OFF: everything on = the whole mix chopped after the effects. **SL PITCH**: the repeats' 16-step pitch (±24 st, tape style, QNT SCL to the scale).

DELAY · CHORUS

DLY

TIME	FDBK	TONE	MIX

DLY 2

WIDE	PITCH	SPRY	MOD

DLY 3

TYPE	PING	DRIVE	R TIME

Synced (4BAR … 1/32; V10.8 dotted 1/4D 1/8D 1/16D) or free (5 ms – 2.95 s; past 0.74 s darker); PITCH / SPRY: a grain delay. TYPE ANALOG / DIGITAL, PING-PONG, DRIVE the loop. **R TIME** (V10.49): the right echo's time as a ratio of TIME (=L, x1/2, x2/3, x3/4, x4/3, x3/2, x2: 1/8 with x3/2 is 3/16); not with PING or the grains.

CHORUS

RATE	DEPTH	STER	-

STER: 0 as always, down to mono, up to twice as wide.

REVERB · FIVE MODELS

TYPE	SIZE	DAMP	PRE
MOD	RATE	FILT	WIDE

TYPE	WHAT IT IS	SIZE	DAMP	MOD	RATE	WIDE
ROOM	Classic room	decay	darkens	chorus on the tail	its speed	stereo
SPRING	Amp spring tank	length · decay	darkens	wobble	wobble speed	stereo
SHIMR	Up an octave on every pass; **PRE → AIR** (±24 st: down darker and lower, up airier and higher)	decay	darkens	how much shimmer	—	—
RESO	4 strings tuned to the scale; **PRE → PITCH** (0–14 steps up), **DAMP → FOLW** (OFF / T1–T6: the chord follows that track's notes)	how long they ring	FOLW	past half: the chord opens	—	—
CLOUD	Granular wash; SIZE max **freezes** it; **PRE → AIR** (±24 st: the cloud down / up, darker / airier)	decay · freeze	darkens	grains pitched + chorus	grain density	grain spread

MASTER

CLIP	PNCH	DUCK	REL

CLIP saturates without getting louder. **PNCH** drives the drums: denser body. **DUCK**: the kick pulls the rest down (REL its release). An always-on **leveler** (slow 2:1, –9…+6 dB) before the limiter keeps patches at a steady volume.

TAPE

TAPE	WOW	-	-

After MASTER: TAPE saturates, darkens the highs and lifts the lows; WOW the wow and flutter (par9: much stronger at 100).

EQ · 4 BANDS (par9)

LOW	LO MID	HI MID	HIGH

–12…+12 dB, wide and gentle (~200 Hz, 1 kHz, 4 kHz); at 0 it costs nothing. The page draws the curve (SCOPE ON: the master live behind it); EQ and TAPE change only when a project loads.

PUNCH-IN FX (FX HELD)

REPEAT 1/8 · 1/16 · 1/32 (1/8 from 41 BPM, 1/16 from 21: V10.59), LPF, HPF, LOOP 3 / 5 / 7, GLITCH on the grid; on the notes: OCT–/+, ½ TEMPO, DEC–/+, STUTTER, ATK+, ARP, RANDOM. Knobs: FILTER CRUSH THROW DEPTH. Recordable per section.

GPL-3.0 · fork of Felucca (Hügelton Instruments) · song mode from SLOOP